Lesson 4: Project: Quiz Show

Module 1 | 60 minutes | score variables and decisions | Grades 6–10 · ages 11–15

Objective: Build a five-question quiz game that keeps score, gives feedback on every answer and awards a final grade.

Success criteria

- I can keep score with a variable and +=.
- I can check answers with if / else and give feedback.
- I can accept answers in any capitals with .lower() and .strip().
- I can give a final grade with if / elif / else in the right order.

Key vocabulary

- **score variable**: A variable that counts points during a game.
- **+=**: Adds to a variable: score += 1 means score = score + 1.
- **.lower()**: Turns text into small letters.
- **.strip()**: Removes extra spaces from the start and end of text.
- **feedback**: A message that tells the player how they did.
- **pattern**: A piece of code you can copy and reuse with small changes.

You will need

- Projector with Kitabash (lesson video, Python Lab)
- Devices: 1 per learner, with a keyboard
- Printed worksheet (optional)
- Printed project brief
- Project files: one_question.py

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game: Human Quiz Machine**	Play Human Quiz Machine (unplugged). The class tries tricky answers like “MaRs”. Discuss: should they count?
5 min	**Hook, video and plan**	Show the hook, play the project video with captions. Thumbs poll: “Who knows their quiz topic already?” Learners write 5 questions and answers on paper.
8 min	**Live demo and build-along**	Run the live demo, then everyone builds “one perfect question” and tests it with MARS, mars and Venus.
5 min	**Predict and plan**	Predict-the-output and the ordering activity. Quick competition: first to the correct order wins a point for their team.

18 min	**Build the Quiz Show**	Learners build the full Quiz Show. Halfway check-in: everyone has at least two questions working. Teacher visits breakout rooms.
6 min	**Debug race**	Race to fix “the unfair grade calculator”. The winner explains why order matters.
9 min	**Quiz swap and show-and-tell**	Pairs in breakout rooms play each other’s quiz and give two stars and a wish. Back together, two learners host their quiz live for the whole class.
4 min	**Submit and exit ticket**	Learners submit the project with a reflection. Exit ticket on the grade order question.

Unplugged warm-up

Human Quiz Machine (5 min game): one learner is the “program”. They ask a quiz question, hold up a scorecard starting at 0, and must say exactly “Correct! +1” or “Not quite. It is ...” and update the card. The class tries to trick them with answers like “MaRs” and “ mars ” — should those count? Leads into .lower() and .strip().

Teaching the idea

Today learners combine everything from Module 1 into one complete game. A score variable starts at 0 and grows with score += 1 when an answer is right. Each question is the same pattern: input, clean with .strip().lower(), if/else with feedback. The final grade uses if/elif/else, and the order matters because Python stops at the first True condition. Encourage learners to build and test ONE question before copying the pattern.

Analogy: The score variable is like the runs board at a cricket ground: it starts at 0, the scorer adds runs after every ball, and you read the total at the end of the innings.

Live demo: 1) Type score = 0 and print(score); run it. 2) Add score += 1 twice and run: ask what changed. 3) Type answer = input("Which planet is called the Red Planet? ") then if answer == "mars": print("Correct!"). Run it and type MARS: point out that nothing prints. 4) Add answer = answer.strip().lower() under the input, run again with MARS: now it works. 5) Add the else and the score line. 6) Type the grade block with if score >= 1 first, set score = 5, run, and show the wrong grade; then reorder to check == 5 first.

Questions to ask

Ask	Listen for
Why must score = 0 come before the first question?	So the score exists and starts at zero before we add to it.
What does score += 1 mean?	Add one to score.
Why use .lower() on the answer?	So MARS, Mars and mars all match "mars".
Why check score == 5 before score >= 3?	Python stops at the first True test; 5 is also >= 3.

How can you test your quiz quickly?	Run it once with all right answers and once with all wrong answers.

Differentiation

Support
Provide the one-question file (download) so learners copy-paste the pattern and only change the question and answer. Allow 3 questions as a first goal.

Stretch
Add a bonus round: ask if they want a bonus question with yes/no, and give 2 points for it. Or show the percentage with score * 100 // 5.

Common misconceptions

Misconception	How to fix it
score + 1 on its own line adds a point	You must store it: score = score + 1 or score += 1.
Putting score = 0 inside each question is fine	That resets the score every time. Put it once at the top.
"Mars" == "mars" is True	Capitals matter. Use .lower() first.
The order of elif checks does not matter	Python stops at the first True one, so check the highest grade first.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Program works	Crashes or fewer than 3 questions run	3–4 questions run; score sometimes wrong	All 5 questions run and the score is always right	Works perfectly and handles odd answers like extra spaces
Decisions and score	No if/else or no score	if/else on some questions	if/else on every question and a correct final grade	Grade order is correct and explained; adds a bonus or percentage
Player experience	No feedback to the player	Says right or wrong only	Kind feedback that shows the right answer	Fun title screen, clear questions and a celebration at the end
Code quality and testing	Hard to read; never tested	Some comments; tested once	Comments for each question; tested with	Neat, well named, and tested by another

			right and wrong answers	player with changes made

Exit ticket
A player scores 5. Your grade code checks score >= 3 first, then score == 5. Which grade do they get, and how do you fix it?

Homework
Host your Quiz Show for two people at home. Write down their scores and one question they found too easy, then make it harder.

Answer key — Kitabash activities

1. Python Lab: one perfect question (perform · python lab)

Goals checked: MARS scores a point · “ mars ” with spaces scores a point · Venus scores nothing · Says Correct for a right answer · Uses if and else

One solution:

```
# One quiz question
score = 0
answer = input("Which planet is called the Red Planet? ")
answer = answer.strip().lower()

if answer == "mars":
    print("Correct! +1 point")
    score = score + 1
else:
    print("Not quite. The answer is Mars.")

print(f"Score: {score}")
```

2. Predict the output: score keeper (perform · code output)

3. Plan the quiz (perform · ordering)

Order: Set score = 0 → Ask a question with input() → Clean the answer with .strip().lower() → Check it with if / else and add 1 if right → Repeat for every question → Print the final score and grade

4. Project: Quiz Show (perform · python lab)

Goals checked: Runs for 5 answers · Wrong answers score 0 · Shows the final score · Uses if, elif and else

Example program:

```
# Quiz Show: India and the World
# Made by: Riya

score = 0
print("=== WELCOME TO THE KITABASH QUIZ SHOW ===")
print("5 questions. 1 point each. Good luck!")
print()

# Question 1
answer = input("1. How many players are in a cricket team? ")
if answer.strip() == "11":
    print("Correct!")
    score += 1
else:
    print("Not quite. It is 11.")

# Question 2
answer = input("2. Which planet is called the Red Planet? ")
if answer.strip().lower() == "mars":
    print("Correct!")
    score += 1
else:
    print("Not quite. It is Mars.")

# Question 3
answer = input("3. What is 12 x 12? ")
if answer.strip() == "144":
    print("Correct!")
    score += 1
else:
    print("Not quite. It is 144.")

# Question 4
answer = input("4. Which festival is called the festival of lights? ")
if answer.strip().lower() == "diwali":
    print("Correct!")
    score += 1
else:
    print("Not quite. It is Diwali.")

# Question 5
answer = input("5. Which animal is India's national animal? ")
if answer.strip().lower() == "tiger":
    print("Correct!")
    score += 1
else:
    print("Not quite. It is the tiger.")

print()
print(f"Final score: {score} out of 5")
if score == 5:
```

```
    print("Grade: Quiz Champion!")
elif score >= 3:
    print("Grade: Star Player")
elif score >= 1:
    print("Grade: Good Try")
else:
    print("Grade: Keep Practising")
```

5. Debug it: the unfair grade calculator (probe · python lab)

Goals checked: 5 gets Quiz Champion · 4 gets Star · 2 gets Good try · 0 gets Keep practising

One solution:

```
score = int(input("Score out of 5? "))
if score == 5:
    print("Grade: Quiz Champion")
elif score >= 4:
    print("Grade: Star")
elif score >= 2:
    print("Grade: Good try")
else:
    print("Grade: Keep practising")
```

Answer key — worksheet

1. **Key words:** score variable = A variable that counts points during a game. · += = Adds to a variable: score += 1 means score = score + 1. · .lower() = Turns text into small letters. · .strip() = Removes extra spaces from the start and end of text.
2. **Plan your program:** Goals: MARS scores a point; " mars " with spaces scores a point; Venus scores nothing; Says Correct for a right answer; Uses if and else. One solution: # One quiz question score = 0 answer = input("Which planet is called the Red Planet? ") answer = answer.strip().lower() if answer == "mars": print("Correct! +1 point") score = score + 1 else: print("Not quite. The answer is Mars.") print(f"Score: {score}")
3. **Predict the output: score keeper:** A. Score: 4
4. **Plan the quiz:** 1 Set score = 0 · 2 Ask a question with input() · 3 Clean the answer with .strip().lower() · 4 Check it with if / else and add 1 if right · 5 Repeat for every question · 6 Print the final score and grade
5. **Plan your program:** Goals: 5 gets Quiz Champion; 4 gets Star; 2 gets Good try; 0 gets Keep practising. One solution: score = int(input("Score out of 5? ")) if score == 5: print("Grade: Quiz Champion") elif score >= 4: print("Grade: Star") elif score >= 2: print("Grade: Good try") else: print("Grade: Keep practising")
6. **My plan:** A clear plan that covers every requirement.
7. **Checklist:** All requirements met.
8. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum.